

Lecture 11

Random and Mixed Effects Models, Computer Experiments

Reading: Oehlert 2010 Chapter 11.1-11.7; DAE 2017 Chapters
17.1-17.3; 17.7& 20

STAT 8020 Statistical Methods II

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Agenda

Random and Mixed
Effects Models,
Computer
Experiments



Random and Mixed
Effects Models

Computer Experiments

1 Random and Mixed Effects Models

2 Computer Experiments

Fixed Effects

Most settings we have dealt with so far (except the analysis of split-plot designs) have involved **fixed effects**:

CRD:	$y_{ij} = \mu + \alpha_i + \epsilon_{ij}$
RCBD:	$y_{ij} = \mu + \alpha_i + \beta_j + \epsilon_{ij}$
Factorial:	$y_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ij}$

- The treatment effects are unknown but constants

Interpretation

The levels of interest are fixed and reproducible.
If we rerun the experiment, we study the same treatments again

- We can increase the power of all of our tests by increasing the sample size n
- We perform inference on the treatment effects via t -tests and F -tests

Random effects models look very similar to fixed effects models. For example, we could have

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}.$$

The difference is in the **assumptions** we make for the treatment effects

Fixed Effects	Random Effects
α_i : unknown constants	$\alpha_i \sim N(0, \sigma_\alpha^2)$
Inference:	Inference:
$\alpha_1, \dots, \alpha_g$	σ_α^2

How and Why Are Things So Different?

Fixed effects:

- The treatments are the treatments and they are unchanging
- If we rerun the experiment, we are still studying the same treatments

Random effects:

- The treatments are a random sample from a population of potential treatments
- If we rerun the experiment, we are looking at an entirely new sample of treatments
- Inference is on the population of potential treatments

Example

A company manufactures thousands of machine parts. We sample 10 machines at random to estimate machine-to-machine variability.

Fixed effects:

- $\text{Var}(y_{ij}) = \sigma^2$
- All y_{ij} s are independent of each other
- Interest is about α_i s

Random effects:

- $\text{Var}(y_{ij}) = \sigma_\alpha^2 + \sigma^2$
- $\text{Cor}(y_{ij}, y_{kl}) = \begin{cases} 0 & \text{if } i \neq k \\ \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma^2} & \text{if } i = k; j \neq l \\ 1 & \text{if } i = k; j = l \end{cases}$
- Interest is (mostly) about σ_α^2

$$H_0 : \sigma_\alpha^2 = 0$$

No between-group variability

An Example of Fixed Effects vs Random Effects

- 1 Compare reading ability of 10 2nd grade classes in NY:

Select $g = 10$ specific classes of interest. Randomly choose n students from each classroom. Want to compare α_i s (class-specific effects) $\Rightarrow$ **Fixed effects**

- 2 Compare variability among all 2nd grade classes in NY:

Randomly choose $g = 10$ classes from large number of classes. Randomly choose n students from each classroom. Want to assess σ_α^2 (class to class variability) $\Rightarrow$ **Random effects**

When Are Random Effects Useful?

- Repeated measurements
- Batch-to-batch variability
- Subject-specific heterogeneity
- Multi-level / hierarchical data
- Sampling from a larger population

Examples

schools, hospitals, manufacturing batches, patients,
classrooms, counties

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij},$$

where

- μ is the overall mean
- α_i : i th treatment effect and $\alpha_i \sim N(0, \sigma_\alpha^2)$
- $\{\alpha_i\}$ and $\{\epsilon_{ij}\}$ independent
- The hypotheses are:

$$H_0 : \sigma_\alpha^2 = 0$$

$$H_a : \sigma_\alpha^2 > 0$$

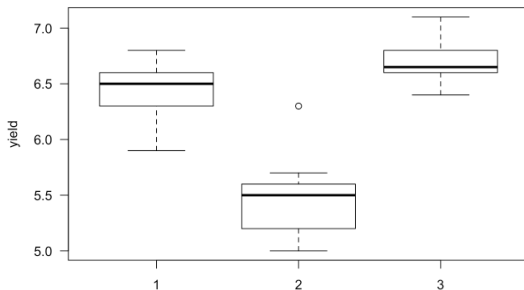
One can use either “old school” method (ANOVA) or “**new school**” method (REML) to make inference about σ_α^2

REML is now generally preferred because it produces more stable and less biased variance component estimates, especially in complex or unbalanced designs

Random Effects Example

Suppose that an agronomist is studying a large number of varieties of soybeans for yield. The agronomist randomly selects three varieties, and then randomly assigns each of those varieties to 10 of the 30 available plots

Soybean	Yield
V1	6.6, 6.4, 5.9, 6.6, 6.2, 6.7, 6.3, 6.5, 6.5, 6.8
V2	5.6, 5.2, 5.3, 5.1, 5.7, 5.6, 5.6, 6.3, 5.0, 5.4
V3	6.9, 7.1, 6.4, 6.7, 6.5, 6.6, 6.6, 6.6, 6.8, 6.8



```
> fixef <- lm(yield ~ var)
```

```
> anova(fixef)
```

Analysis of Variance Table

Response: yield

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
var	2	8.306	4.1530	49.593	9.114e-10 ***
Residuals	27	2.261	0.0837		

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> coefficients(fixef)
```

(Intercept)	var2	var3
6.45	-0.97	0.25

Random Effects Analysis

```
> library(lme4)
> randef <- lmer(yield ~ 1 + (1|var), REML = TRUE)
> summary(mod1)
```

Linear mixed model fit by maximum likelihood . t-tests
use Satterthwaite's method [lmerModLmerTest]
Formula: yield ~ 1 + (1 | var)

AIC	BIC	logLik	deviance	df.resid
27.2	31.4	-10.6	21.2	27

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.8755	-0.6033	0.1245	0.5068	2.7574

Random effects:

Groups	Name	Variance	Std.Dev.
var	(Intercept)	0.26849	0.5182
Residual		0.08374	0.2894

Number of obs: 30, groups: var, 3

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	6.2100	0.3038	3.0000	20.44	0.000256

Concrete Cylinder Example Revisited

Suppose you are manufacturing concrete cylinders for bridge supports. There are three ways of drying concrete (say A, B, and C), and you want to find the one that gives you the best compressive strength. The concrete is mixed in batches that are large enough to produce exactly three cylinders, and your production engineer believes that there is substantial variation in the quality of the concrete from batch to batch.

Treatment	Batch					Trt Sum
	1	2	3	4	5	
A	52	47	44	51	42	236
B	60	55	49	52	43	259
C	56	48	45	44	38	231
Batch Mean	168	150	138	147	123	726

If we instead treat the batch effects as random effects, then we obtain a **Mixed Effects Model**, since the model now contains both:

- **fixed effects** (treatment effects), and
- **random effects** (batch effects)

Concrete Cylinder Example: Mixed Effects Analysis

```
> randef <- lmer(x ~ trt + (1|blk), REML = TRUE, data = dat)
> summary(randef)
```

Linear mixed model fit by REML. t-tests use

Satterthwaite's method [lmerModLmerTest]

Formula: x ~ trt + (1 | blk)

Data: dat

REML criterion at convergence: 71.1

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.1417	-0.6147	-0.1494	0.5772	1.3390

Random effects:

Groups	Name	Variance	Std.Dev.
blk	(Intercept)	28.35	5.324
Residual		5.85	2.419

Number of obs: 15, groups: blk, 5

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	47.200	2.615	5.054	18.047	8.76e-06
trtB	4.600	1.530	8.000	3.007	0.0169
trtC	-1.000	1.530	8.000	-0.654	0.5316

Motivation: Why Use Computer Models?

Real-world **physical experiments** may be:

- too **costly**, **time-consuming**, or **dangerous**
- **ethically prohibited** or **physically infeasible**

Computer simulations allow us to study complex systems virtually before conducting expensive real-world experiments.

Applications:

- Climate and hurricane forecasting
- Aircraft crash simulations
- Engineering and industrial design

Examples of Computer Models

Random and Mixed
Effects Models,
Computer
Experiments

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Random and Mixed
Effects Models

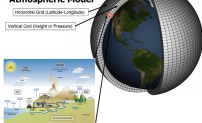
Computer Experiments

MATH MODEL
(GOVERNING EQUATIONS)

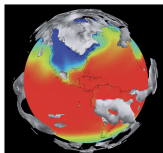
← approximates

COMPUTER MODEL
(DISCRETIZATION)

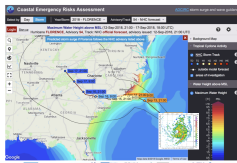
Schematic for Global
Atmospheric Model



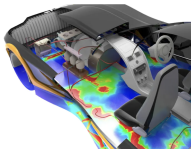
Source: Wikipedia: Climate Model



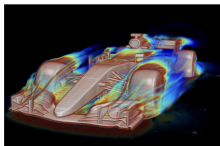
Source: Pacific Northwest National Laboratory



Source: Coastal Emergency Risk Assessment



Source: Ansys, Inc.

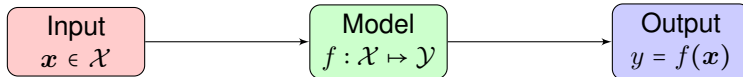


Source: Simcenter STAR-CCM+



Source: MATLAB & Simulink

- The physical process can be described by a mathematical model (e.g., a system of differential equations)
- Computer code (simulator) can be written to compute the response from the mathematical model



Computer Model: $y = f(x)$

- f is typically **not available in closed form**
- Constructed from scientific or engineering principles
 $\Rightarrow$ “law-driven” (mechanistic) model
- Often **deterministic** e.g., numerical solutions of PDEs
- May not have explicit observed data $\{x, y\}$

Computer Experiment:

- A computer simulator defines an input-output relationship:

$$y = f(x)$$

- A **model run** consists of evaluating the simulator at a chosen input setting x
- A computer experiment involves performing many model runs over different input settings

Computer Experiments vs. Physical Experiments

- *“Experimental results are believed by everyone, except for the person who ran the experiment”*
- *“Computational results are believed by no one, except the person who wrote the code”*

Replication, randomization and blocking are mostly irrelevant for a computer experiment because many **computer codes are deterministic** and **all the inputs to the code are known and can be controlled**

Key Difference

Physical experiments: uncertainty comes from observational noise

Computer experiments: uncertainty comes from limited simulator runs

Design and Analysis of Computer Experiments

A computer experiment evaluates a simulator:

$$y = f(\mathbf{x})$$

at selected input settings:

$$\{\mathbf{x}_i\}_{i=1}^n$$

Two major statistical tasks:

- **Design:** Choose which input configurations

$$\{\mathbf{x}_i\}_{i=1}^n$$

to run the simulator

- **Analysis:** Use observed data

$$\{(\mathbf{x}_i, y_i)\}_{i=1}^n$$

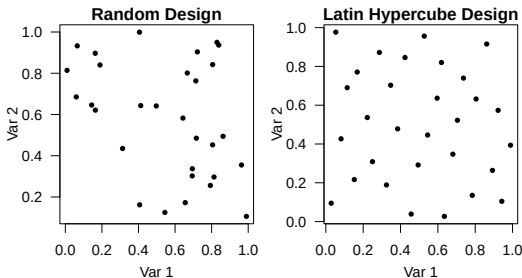
to estimate the input-output relationship

$$y = f(\mathbf{x})$$

Design of Computer Experiments

Question: where to make the runs, i.e., the selection of inputs $\{x_i\}_{i=1}^n$ for a given computational budget n .

Example: $x_i = (x_{i1}, x_{i2})^T \in [0, 1]^2$ with $n = 30$



Space-Filling Designs

Space-filling designs spread runs throughout the input space to learn the global behavior of the simulator.

Example: [Latin hypercube designs \(LHDs\)](#)

Classical DOE vs Computer Experiment DOE

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Random and Mixed
Effects Models

Computer Experiments

Physical Experiments	Computer Experiments
Focus on noise reduction to gain statistical efficiency	Focus on space-filling designs
Replication is often important	Replication is often unnecessary
Factorial and blocking designs	Latin hypercube and uniform designs
Estimate treatment effects	Learn the global response surface

Why Build Statistical Emulators?

Goal

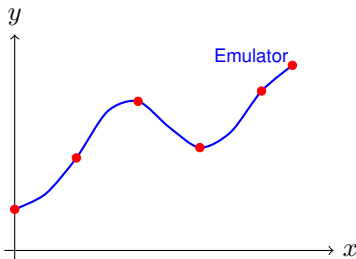
Build a fast emulator to learn

$$y = f(x)$$

using only a small number of expensive simulator runs

Running a complex simulator may require:

- hours or days of computation
- large-scale computing resources
- expensive supercomputer runs



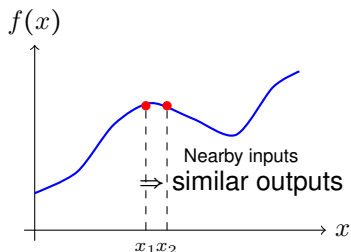
The emulator approximates the simulator using only selected runs

Using Gaussian Processes for Emulator Modeling

Key Assumption

Nearby inputs $\Rightarrow$ similar outputs

- Computer simulators often produce **smooth responses**
- Small changes in the input usually lead to small changes in the output



GP Emulator

A **Gaussian process (GP)** emulator models this smooth behavior using correlation across the input space

Gaussian Process (GP) Emulation

Given simulator runs $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, we model the unknown $y = f(\mathbf{x})$ using a **Gaussian process (GP)**:

$$f(\mathbf{x}) \sim GP(m(\mathbf{x}), K(\mathbf{x}, \mathbf{x}'))$$

- **Mean function:** $m(\mathbf{x}) = \mathbb{E}[y(\mathbf{x})]$, often chosen to be simple, e.g.

$$m(\mathbf{x}) = \mu$$

- **Covariance function:** $K(\mathbf{x}, \mathbf{x}') = \text{Cov}(y(\mathbf{x}), y(\mathbf{x}'))$. A common choice is a separable covariance:

$$K(\mathbf{x}, \mathbf{x}') = \sigma^2 \prod_{j=1}^p C_j(d(\mathbf{x}_j, \mathbf{x}'_j); \theta_j)$$

which models correlation through componentwise distances across inputs

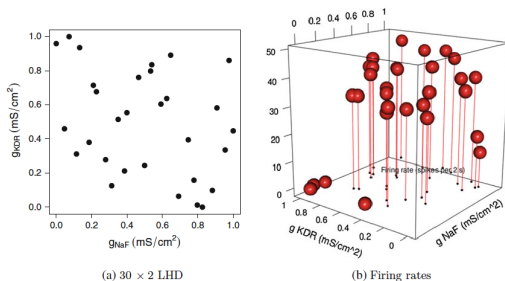
Key Tasks

Estimate GP parameters using **maximum likelihood**, then use the fitted GP for prediction and uncertainty quantification

Neuron Experiment [pp.776-778, DAE 2017]

The firing rate of a neuron at +380 pA current injection of a young monkey is modeled as a deterministic function of two input variables $x = (g_{NaF}, g_{KDR})$:

- g_{NaF} : maximal conductance of the transient sodium
- g_{KDR} : maximal conductance of the delayed-rectifier potassium



Source: Fig. 20.6, DAE 2017

Goal: predict firing rates at untried input settings

Neuron Experiment Result

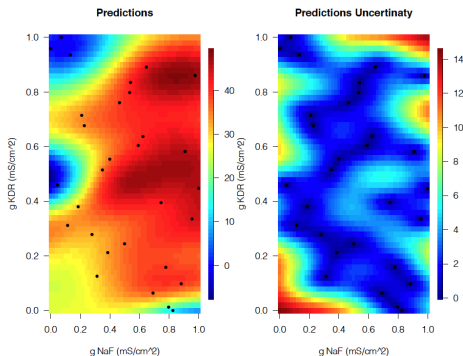
- GP fitted using the squared exponential covariance:

$$C(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp[-\theta_1(x_1 - x'_1)^2 - \theta_2(x_2 - x'_2)^2]$$

- Estimated parameters:

$$\hat{\mu} = 27.61, \quad \hat{\sigma}^2 = 251.86, \quad \hat{\theta}_{\text{NaF}} = 5.03, \quad \hat{\theta}_{\text{KDR}} = 50.22$$

- GP predictions (**Left**) and prediction uncertainties (**Right**)



Interpretation

Prediction uncertainty is smallest near observed runs and largest in sparsely sampled regions.

These slides cover:

- Random and Mixed Effects Models
- Computer Experiments: Concepts, Design and Analysis

R functions to know:

- **Random and Mixed Effects Modeling:** `lmer` from the packages `lme4/lmerTest`
- **Design and Analysis of Computer Experiments:** `maximinLHS` from the package `lhs` for conducting Latin hypercube sampling designs and `mlegp` from the package of the same name for GP emulation